GENERATIVE AI INTERNSHIP TRAINING CURRICULUM

Training Schedule and Learning Objectives

The 30-day internship program is structured into six progressive training phases, designed to help students develop a strong foundation in Generative AI and Large Language Models, gain practical AI application development experience, and build a functional AI-powered project by the end of the program.

Duration: 30 Days
Technology Stack: Python, Streamlit, ChromaDB, Git & GitHub

Days	Training Phase	Topics Covered
Day 1–5	Generative AI & LLM Fundamentals	Introduction to Generative AI, AI Agents and Agentic AI; how LLMs work, including tokens, embeddings, attention and temperature; API keys and environment setup; first API calls; messages and roles (system, user and assistant).
Day 6–10	Prompt Engineering & Structured Output	Zero-shot and few-shot prompting, chain-of-thought concepts, role and tone control, prompt templates, reliable JSON output, error handling and building web interfaces with Streamlit.
Day 11–15	Embeddings, Vector Databases & Semantic Search	Text embeddings, representing meaning as vectors, semantic vs. keyword search, vector databases (ChromaDB), storing and querying documents, and building a semantic search engine.
Day 16–20	Retrieval-Augmented Generation (RAG)	RAG pipeline, document chunking, retrieval and context injection, hallucination reduction techniques, source citations, persistence and multi-document knowledge assistants.
Day 21–25	Tool Calling, AI Agents & Agentic AI	Function and tool calling, tool schemas, tool-calling workflows, the agent loop (think–act–observe), agent memory and planning, multi-step autonomous agents and safety limits.

Day 26–30	Final Project Development & Evaluation	Project planning, requirement analysis, evaluation and guardrails, cost control, deployment preparation, testing, debugging, documentation, deployment and final project presentation.

Expected Learning Outcomes

Upon successful completion of the internship, students will be able to:

1. Explain how Large Language Models work and interact with them through APIs.
2. Apply prompt engineering techniques to control and improve AI output quality.
3. Produce reliable structured (JSON) output and build AI-powered web applications using Streamlit.
4. Create embeddings and implement semantic search using vector databases.
5. Build Retrieval-Augmented Generation (RAG) applications that answer questions from custom documents with citations.
6. Implement tool/function calling and develop autonomous AI agents that use tools in a loop.
7. Apply fundamental AI safety, evaluation, and cost-control practices.
8. Develop, test, deploy, and present a functional Generative AI application.

Training Methodology: Concept-based learning, instructor-led demonstrations, hands-on coding sessions, daily practical exercises, assignments and project-based learning.

Program Completion Deliverable: A functional Generative AI application (such as a RAG-based document assistant or an AI agent) developed, deployed, and presented by the student.